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Dynamic assessment of L2 argumentative writing: Understanding the effectiveness of ZPD-based instructional enrichment

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Abstract

Argumentative writing has long been recognized as challenging for English second language (L2) writers yet central to their academic success in educational settings. This article investigates the effectiveness of a writing instructional enrichment program informed by diagnoses from dynamic assessment (DA) in promoting L2 learners' argumentative writing abilities. Grounded in Vygotskian Sociocultural Theory, DA introduces mediation into the assessment procedure, so that learner emerging abilities are diagnosed through observing their responsiveness to mediation when difficulties arise. In this study, prior to and following the instructional programs, a three-step, interactionist DA procedure was implemented where participants composed argumentative essays in response to reading–writing integrated tasks. Following the initial DA procedure, participants were assigned to either an enrichment program that received individualized instruction on integrated argumentative writing targeting their Zone of Proximal Development (ZPD), or a non-enrichment program, where generic and standard instruction uninformed by DA diagnoses was provided. Finally, the study implemented a transfer assessment to ascertain the participants' ability to transfer their learning to a more challenging, complex integrated task. The essays completed independently by both enrichment and non-enrichment groups during the assessments were analysed in terms of stance taking and stance support. Stance taking was examined through writers' establishment of their own position and their engagement with opposing positions, whereas stance support was investigated in four sub-aspects: type of

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evidence, number of source ideas, non-transgressive/transgressive intertextuality, and content accuracy. The findings revealed that the enrichment group outperformed the non-enrichment group in stance taking and stance support at both the DA procedure implemented following the writing instructional programs and the transfer assessment. The article concludes by highlighting the effectiveness of tailoring writing instruction towards learners' ZPD as diagnosed through DA in promoting their development of integrated argumentative writing.

Keywords
dynamic assessment, instructional enrichment, integrated argumentative writing, Vygotskian sociocultural theory, Zone of Proximal Development

1 Introduction

The importance of argumentation in academic contexts for English language learners has been well documented in the field of second language (L2) writing (e.g. Lee & Deakin, 2016; Pessoa et al., 2017; Plakans & Gebril, 2017). Students engage in argumentative writing in both English academic writing courses as well as across disciplines and subject areas (Hirvela, 2017). Argumentation also features in high-stakes, large-scale standardized tests of English, including IELTS (International English Language Testing System) and TOEFL (Test of English as a Foreign Language). At the same time, the argumentative genre of writing has been shown to be particularly challenging for English L2 writers (Bacha, 2010; Miller & Pessoa, 2016; Pessoa, 2017; Zhu, 2001).

The research on L2 argumentation has examined writing processes and cognitive load involved when writers engage in argumentative writing tasks (Li & Wang, 2024; Vandermeulen et al., 2024), students' social interaction and collaborative argumentation in online and face-to-face settings (Jin et al., 2020; Su et al., 2021; Wu et al., 2023), and the structural components and quality of L2 learners' argumentation as analysed through Toulmin's ([1958] 2003) model of argumentation or its modified forms (e.g. Abdollahzadeh et al., 2017; Liu & Stapleton, 2014, 2020; Qin, 2013; Qin & Karabacak, 2010; Stapleton & Wu, 2015). In particular, recent years have seen an upsurge in studies on L2 argumentative writing that address a wide range of issues, including linguistic features of argumentative texts, the instruction and assessment of L2 writers' argumentation, and strategies adopted by students in producing argumentative essays (Farsani et al., 2025).

Despite ongoing interest in L2 argumentation, it has also been noted that while argumentative writing in academic contexts normally requires use of source materials, and in L2 assessment, reading–writing integrated tasks are often employed to assess students' argumentative writing, few studies on L2 argumentation have examined student efforts to integrate sources into their writing (Al Bulushi, 2022; Chuang & Yan, 2023). Writing from an L2 assessment perspective, Plakans and Gebril (2017, p. 86) observed that 'argumentative writing may in fact suffer from a lack of theoretical and empirical attention.' Similarly, Chuang and Yan (2022) point out that although argumentation is the genre widely used to assess academic writing, 'little is known about the characteristics of argumentation in writing assessment contexts' (p. 3). Noting that the genre is assessed

not only in large-scale standardized tests but also by writing instructors in classrooms, Plakans and Gebril (2017) call for a coherent alignment of approaches to teaching and assessing learner L2 argumentative writing development.

This article responds to this call by reporting a project that employed insights into learner argumentative writing abilities obtained through dynamic assessment (DA) to inform an instructional enrichment program. DA is grounded in Vygotsky's (2012) Sociocultural Theory (SCT), according to which learner independent performance of assessment tasks reveals only their ability to self-regulate, that is, the products of their development up to the present time; the aim of DA is to expand this diagnosis to include abilities that have not yet fully developed but that are still emerging (Poehner, 2008). This range of abilities, which Vygotsky termed the Zone of Proximal Development (ZPD), are apprehended during assessment by introducing forms of support, or *mediation*, when learners encounter difficulties. Learner responsiveness to mediation (e.g. identifying errors and resolving them) and the quality of mediation they require (e.g. reminders, feedback, models) reveal whether abilities have begun to emerge and how close they are to full development (Poehner & Lantolf, 2024). The importance of this for education, according to Vygotsky (2011), is that abilities that are in the ZPD provide the ideal focus for instruction. In other words, instruction should not be directed at abilities that have already developed but rather at abilities that are partly formed and that can be identified by learner responsiveness to mediation during DA. DA then provides a diagnosis of learner ability that serves to orient instruction to have the greatest impact on learner development.

The data reported in this article were taken from a larger project that implemented DA with 13 L2 English learners recruited from a large U.S. university (for full details of the project, see Yu, 2023). In the study, students were divided into an enrichment group, which received individualized argumentative writing instruction informed by the results of their initial DA, and a non-enrichment group (control group), which continued to receive 'standard' writing instruction that did not attempt to take account of the DA results. Put another way, instruction for the enrichment group was oriented to each learner's ZPD whereas instruction for the non-enrichment group continued to follow a set curriculum (described in detail later). Instruction for both groups entailed five weekly one-to-one sessions with an experienced writing teacher (also one of the authors). Following this phase of the study, a parallel version of the initial DA procedure was administered to permit tracking of both individual learner development over time and between-group comparisons. Finally, the study included what DA researchers refer to as a *transfer assessment*, that is, assessment in which tasks have been designed to provide a higher degree of challenge to learners in order to observe how well they maintain any gains that have appeared following an instructional intervention. The idea of transfer is conceptually rooted in the work of Israeli developmental psychologist Reuven Feuerstein and the notion that true development entails the ability to move beyond the immediate goals of a here-and-now situation to recontextualize the acquired cognitive functions (e.g. Feuerstein et al., 2002, 2010).

Full discussion of this project is beyond the scope of the present article (readers are referred to Yu, 2023). This article takes a qualitative approach and focuses specifically on two defining components of argumentative practices: stance taking and stance

support (Chandrasegaran, 2008; Chandrasegaran & Kong, 2006). The major research questions we explore are:

- Research question 1: How effectively does DA-informed instructional enrichment promote L2 learner development of stance taking in their argumentative writing?
- Research question 2: How effectively does DA-informed instructional enrichment promote L2 learners' integration of information from source materials as well as their own reasoning in elaborating their stance in argumentative writing?

We approach both questions through examination of texts learners independently produced prior to and following their participation in the instructional program to which they were assigned (i.e. enrichment or non-enrichment). Tracing learners across time as well as comparing the performances between the two groups illuminate development that occurred and how those receiving DA-informed enrichment fared relative to those who received standard instruction. We conclude with a discussion of implications for L2 writing instruction and assessment.

II Literature review

1 Dynamic assessment and use of the ZPD to orient instruction

Barrs (2022) traces Vygotsky's discussions of the ZPD and its relevance to both assessing and teaching to his efforts to understand how education could meet the needs of all learners during a period of rapid expansion of public schooling in Russia and the influx of children from diverse linguistic and cultural backgrounds. Standard measures of ability, such as IQ, frequently yielded low scores for all these groups, resulting in their placement in remedial or special education programs. As Barrs explains, Vygotsky proposed to shift the focus of assessment so that the driving question was no longer, what are learners able to do? but rather, how successfully can they go beyond their current abilities? Procedurally, the new assessment Vygotsky devised involved the use of hints, reminders, leading questions, and models when learners encountered difficulties and appeared unable to continue. The outcome of the procedure was the observation that individuals varied with regard to their responsiveness to such support, referred to as mediation.

The diagnosis of abilities that results from this assessment procedure, according to Vygotsky (1998), reveals how instruction can optimally promote the development of learner abilities. He lamented that too often instruction is oriented toward what the learner is perceived as 'ready' to acquire. In his view, waiting for development to occur on its own and then offering instruction is precisely the wrong way to understand the value of education. Instead, he saw instruction as needing to be out in front of development, but not so far ahead that it disregards learner needs. The 'great practical significance' of the ZPD for education is that it provides a target for instruction: the range of abilities that have already begun to emerge but that have not yet fully developed (Vygotsky, 1998, p. 204). It is in this way that Vygotsky (2012, pp. 199–200) submitted that 'the only good kind of instruction is that which marches ahead of development and leads it; it must be aimed not so much at the ripe as at the ripening functions.'

The use of DA to identify 'ripening functions' that can be the focus of instruction has been discussed extensively outside the L2 field, particularly in special education and the development of general cognitive abilities (see Kozulin, 2024). With L2 learners, Poehner (2008) employed DA to diagnose use of verbal tense and aspect among university learners of French during oral narration tasks. The resulting diagnostic profiles informed individualized instructional enrichment in which learners completed various narration tasks of increasing complexity, frequently engaging in dialogic mediation with a language teacher. That study's research design did not include a control group, as the aim was not to compare a ZPD-informed instructional program to other forms of instruction but rather was to document learner developmental processes over time. Since then, a number of studies have implemented DA in contexts of L2 writing.

Rahimi et al. (2015) used DA with advanced L2 learners of English to identify the point during a process approach to writing where learners were experiencing the greatest difficulties: during brainstorming of ideas, generating outlines, or drafting thesis statements for their essays. An implicit-to-explicit scale of mediating prompts was followed so that the researchers could determine precisely how much mediation learners required for each of the three stages of process writing. Kushki, Rahimi, and Davin (2022) focused their DA procedure on task response in learner argumentative writing, a dimension of the IELTS writing rubric. The researchers implemented their DA in an online, asynchronous format that included three rounds of review of student essays via email. During each round, mediation was provided to learners in order for them to have the opportunity to revise their writing. Mediation became more explicit with each round of email review so that researchers could trace improvements that individuals made over time as well as their responsiveness to mediation.

Nassaji and colleagues (Kushki, Nassaji, & Rahimi, 2022; Nassaji et al., 2020) examined English L2 writing abilities among university learners in Iran. Their research advanced prior DA studies on L2 writing by taking a comparative approach to examine the differential effectiveness of the use of scripted, standardized mediation (interventionist DA) and mediation that was open-ended and dialogic (interactionist DA). Adopting a revised version of Toulmin's model of argumentation, the research found the superiority of interactionist DA in diagnosing learners' ZPDs and struggles in argumentative writing. The research conducted by Nassaji and colleagues focused on DA procedures and included repeated administrations of DA, but it did not tie DA diagnoses to subsequent, more sustained writing instruction to provide targeted intervention.

To our knowledge, Yu and Poehner (2023) was the first study where DA served as the basis for individualized instruction targeting L2 writers' ZPD. The study found that DA-informed writing instruction was more conducive than standard writing tutoring in promoting L2 English language learners' academic writing development, as measured by changes in rubric ratings assigned to the learners' essays. While the adopted rubric evaluated the learners' essays in terms of content, organization, language accuracy, and the range and complexity of the language, the specific textual features that contributed to the effectiveness of argumentation were not examined. The current study builds upon that research to investigate the components that are essential to effective integrated argumentative writing: establishing and sustaining a clear position, engagement with counterarguments, and supporting the stance with information from source materials as well as

a writer's own opinions. Moreover, this study tracks learners' writing development further by examining their performance during a transfer assessment.

2 Conceptualizing argumentation in academic writing

One complexity in studying argumentative writing is that students engage in the activity in widely different academic contexts and disciplines. Argumentation is always situated in time and place and varies by academic or pedagogical traditions (e.g. Kibler, 2017; Kibler & Hardigree, 2017). The instruction of argumentative writing in the U.S. secondary and university levels, for example, draws upon widely different pedagogical practices and backgrounds (Kibler, 2017). In addition, rhetorical structures and techniques vary across cultures, and compared with monolingual first language (L1) students, L2 writers are likely to have already learned how to write in both their first language and English by the time they begin their university writing class (e.g. Kirkpatrick, 2017).

Despite varied contexts in which argumentative writing takes place, taking a stance or position and supporting it with evidence and reasons are considered intrinsic to argumentative practices (Chandrasegaran, 2008; Chandrasegaran & Kong, 2006; Farsani et al., 2025; Inch & Warnick, 2010; Liu & Stapleton, 2014). Stapleton (2017, p. 84) explains that 'argumentative writing – whether in the West, East, or places in between – is premised upon claims supported by reasons and evidence.' Similarly, Kibler and Hardigree (2017, p. 77) note that 'a key element of any argumentative text – and what distinguishes it in many ways from other written genres – is the evidence that is used to establish or further a given argument.'

Synthesizing previous definitions of argumentation, including that of Toulmin et al. (1984), Wingate (2012) identified three interconnected components of the concept: (1) establishing a position, (2) presenting the position through a coherent, logical structure, and (3) selecting and using information from sources as evidence to develop the position. Among these, the first component was proposed as the core element of argumentation. An analysis of the collected student essays and tutor comments found that while the tutors commented most frequently on the deficiency with structure in the students' essays, the issue was often linked to deeper difficulties, namely weak positioning or struggle with selecting and evaluating source information (Wingate, 2012).

Related to establishing and sustaining a position, a common challenge encountered by both English L1 speakers and L2 learners is considering opposing or alternative points of view (e.g. Chuang & Yan, 2022; Nussbaum & Kardash, 2005; Qin & Karabacak, 2010; Wolfe & Britt, 2008). The tendency to focus only on one's own position and to exclude other information or arguments was termed 'myside bias' by Wolfe and Britt (2008). According to these authors, if the thesis is obvious or universally accepted, there is no need to develop the argument, but if the thesis is untenable, readers will not likely be persuaded by even the most cogent arguments. In other words, a debatable thesis allows for reasonable arguments on more than one side of the issue. Research conducted in the context of English L2 writing has found that the inclusion of counterarguments and rebuttals enhances the persuasiveness of the essays and is positively correlated with scores assigned to the essays (Liu & Stapleton, 2014; Qin & Karabacak, 2010).

Use of evidence to develop one's position is also crucial to the quality of reasoning. Argumentation in academic contexts is often research-based (i.e. involving source materials), and reading–writing integrated tasks are commonly used in the assessment of L2 argumentative writing (Chuang & Yan, 2023). Nevertheless, research on L2 argumentative writing has rarely examined L2 writers' source use in constructing arguments. Al Bulushi (2022) investigated the sources cited in one particular component in the argumentative essays composed by her students for a writing course: counterarguments. The study found that when writing counterarguments, the participants mostly cited sources that either supported their points of view or provided background information on the topic. No citation in the counterarguments held viewpoints opposing to the ones taken by the students, and the only two opposing-sided sources found in the collected data appeared in the opening paragraphs of the essays to provide general information on the issues under discussion. Chuang and Yan (2023) collected data from an English placement test administered to international students to investigate source use characteristics in the argumentative essays. One major finding of the study was that more sources were used in the essays with simpler structures or less-developed arguments. The authors interpreted this simple increase in source referencing as a strategy by the writers to make their essays longer. They suggested that writing tasks should include source readings providing both supporting and opposing points of view to challenge test-takers to critically engage with source materials. Similarly, Al Bulushi (2022) proposed that writing instructors should expose students to sources that take different perspectives and encourage them to include source information that oppose their own standpoints.

In the present study, reading passages taking different perspectives on a shared topic were included in each writing task presented in the DA procedures. The prompt asked the participants to take a position on the theme of the readings and support their stance with information from the sources as well as their own reasons or experiences. As mentioned, the central question driving the study concerns how DA and a DA-informed enrichment program may promote English L2 learners' integrated argumentative writing in terms of stance taking and stance support. Specifically, stance taking is examined through two features: establishing a position and engagement with counterarguments, whereas stance support is investigated in four aspects: (1) source of evidence; (2) number of ideas from source readings; (3) non-transgressive/transgressive intertextuality (Borg, 2009; Neumann et al., 2019); and (4) accuracy of source use (Neumann et al., 2019; Uludag et al., 2019). More details are provided in the following section.

III Method

1 Participants

Participants in this study include 13 ESL students recruited from English academic writing classes at a large public U.S. university. They were between 18 and 30 years of age and had learned English for six to 13 years at the time of data collection. The students came from different L1 backgrounds: Arabic, Mandarin Chinese, Japanese, and Vietnamese. Specific learner names referred to in this article are pseudonyms. Table 1 presents the biographical profiles of the participants.

Table 1. Biographical profiles of participants.

Name	Gender	Age (years)	L1	Length of time learning English (years)	Length of time living in the U.S. (months)	Instructional grouping
Hassan	Male	18	Arabic	11	1	Enrichment
Amir	Male	18	Arabic	13	48	Enrichment
Abdur	Male	29	Arabic	7	9	Enrichment
Xuan	Male	23	Vietnamese	10	12	Enrichment
Hina	Female	21	Japanese	7	2	Enrichment
Ying	Female	19	Chinese	11	12	Enrichment
Zamir	Male	18	Arabic	12	3	Non-enrichment
Basim	Male	18	Arabic	12	2	Non-enrichment
Kumi	Female	30	Japanese	6	5	Non-enrichment
Jiehua	Male	18	Chinese	12	1	Non-enrichment
Ahmad	Male	18	Arabic	8	9	Non-enrichment
Amelia	Female	18	Chinese	9	2	Non-enrichment
Larry	Male	18	Chinese	12	1	Non-enrichment

At the beginning of the study, all participants indicated that they had not learned English academic writing systematically or received training in source-based writing. Following the initial DA procedure (explained in Section III.2), learners were randomly assigned to either an enrichment program group (hereafter 'enrichment group') to be provided with instruction tailored to the diagnoses obtained from their DA performance, or a non-enrichment program group (hereafter 'non-enrichment group') where the instruction was standard and generic, not taking into account the learners' individual ZPDs. The enrichment group and non-enrichment group comprised six and seven participants respectively. Both groups met with the instructor, or mediator, an experienced academic writing teacher, individually for five weeks of writing instruction. Each session lasted approximately one hour.

2 Data collection

The study was carried out in four stages over 10 weeks: Time 1 assessments; a five-week, one-to-one writing instructional program; Time 2 assessments; and Transfer assessment. At both Time 1 and Time 2 a Sandwich format of DA (Sternberg & Grigorenko, 2002) was adopted that comprised three steps:

1. Each learner completed an argumentative essay independently in response to a reading–writing integrated task;
2. The mediator conducted one-to-one, interactive DA sessions with the learners to review their essays with reference to an analytic rubric for evaluating academic argumentative writing (Weigle, 2004); and
3. The learners revised their essays independently.

Table 2. Data collection procedures.

Week	Enrichment group	Non-enrichment group
1	Time 1 assessment (Step 1): Learner independent draft: Writing task 1	Time 1 assessment (Step 1): Learner independent draft: Writing task 1
2	Time 1 assessment (Step 2): Interactive DA session 1 Time 1 assessment (Step 3): Learner independent revision: Writing task 1	Time 1 assessment (Step 2): Interactive DA session 1 Time 1 assessment (Step 3): Learner independent revision: Writing task 1
3–7	Instruction informed by DA session 1 diagnosis, tailored to individual needs and problems; approximately one hour each week	Standard, generic writing instruction; approximately one hour each week
8	Time 2 assessment (Step 1): Learner independent draft: Writing task 2	Time 2 assessment (Step 1): Learner independent draft: Writing task 2
9	Time 2 assessment (Step 2): Interactive DA session 2 Time 2 assessment (Step 3): Learner independent revision: Writing task 2	Time 2 assessment (Step 2): Interactive DA session 2 Time 2 assessment (Step 3): Learner independent revision: Writing task 2
10	Transfer assessment: Learner independent draft: Writing task 3	Transfer assessment: Learner independent draft: Writing task 3

Prior to the DA sessions, the mediator read through learner essays to determine the issues and areas of writing to be discussed during DA with each individual. In the second step (i.e. the DA session), which took place the following week, each learner was first invited to make revisions to their essay without support from the mediator. The learner and mediator then jointly reviewed the essay as the mediator used prompts, hints, leading questions, suggestions, etc. to address problems in the writing, adhering to the DA principle of moving from implicit forms of mediation toward increasingly explicit mediation (Aljaafreh & Lantolf, 1994). The learners and mediator met in person for all the DA and writing instructional program sessions and carried out each session in a one-to-one format. All the DA interactions and writing instructional program sessions were video- and audio- recorded. During the study, each participant composed altogether five drafts on three writing tasks: two drafts for each of the first two writing tasks, and one draft for the final, transfer assessment. The overall data collection procedures are presented in Table 2.

3 Writing tasks and transfer assessment

The tasks presented at Time 1 and Time 2 assessments were modified versions of the integrated writing tasks developed at Georgia State University and were used in this study with permission. A sample of the tasks, different from the ones in the present study, can be found in Weigle (2004). Each task included at Time 1 and Time 2 presented two pieces of reading holding directly opposing views on a shared topic. At Time 1 the topic

Table 3. Text characteristics of reading passages used in the assessments.

Assessments	Topic	Text	Length (words)	Flesch index of reading ease	Flesch–Kincaid grade level
Time 1 DA	Homework	1	315	40.2	11.7
		2	293	43.0	11.8
Time 2 DA	Gap-year	1	300	45.4	12.2
		2	353	50.6	12.0
Transfer assessment	Homeschooling	1	177	24.1	13.8
		2	273	37.7	13.2
		3	383	50.1	11.8

Source. Adapted from Yu, 2023.

concerned the value of homework and for Time 2 the phenomenon of a gap year prior to entering university.

The transfer assessment was administered non-dynamically in the last stage of the study. The task presented in this final assessment included three passages expressing diverse points of view on the topic of homeschooling. The learners were asked to take a position on the topic and support their position with evidence, as they did in the previous tasks, but were expected to synthesize and integrate information from more complex sources. The total amount of reading included in the transfer assessment was greater than that in the previous two DA sessions, and whereas the first two tests were parallel in difficulty, the last one was more challenging in terms of Flesch readability scores. In each writing task, participants were instructed to take a position and to support their views with specific information from the readings as well as their own reasons or experiences.

The text characteristics of the source readings are summarized in Table 3. Flesch Index of Reading Ease measures a text with a score ranging from 0 to 100; the higher the score, the easier the text. Flesch–Kincaid Grade Level rates a text on a U.S. school grade level, with higher grade level indicating greater difficulty of the text.

4 Writing instruction in enrichment and non-enrichment programs

Following Time 1 assessments, two writing instructional programs were implemented with the participants in a one-to-one format. The instructional foci in the enrichment program were specific to each individual participant as the learners encountered varied difficulties in completing the task. Nevertheless, all participants struggled to address opposing standpoints and to integrate source information in developing their arguments, and so these two areas were included as instructional topics for all enrichment group learners. Given that the enrichment program was ZPD-sensitive (i.e., informed by the diagnoses of individuals obtained through DA at Time 1), the amount of time devoted to these topics varied from individual to individual reflecting the learner's level of understanding and responsiveness to mediation. In addition, other areas were identified as

problematic for some learners but not others (e.g., lack of coherence and unity, non-academic register, inaccurate linguistic forms, etc.).

As an example, when documenting sources was the focus of instruction, enrichment sessions involved the following: explaining reasons for using sources; introducing the concepts of quoting, paraphrasing, and summarizing as major devices for incorporating source materials; and possible steps a writer can take in paraphrasing. Exercises included asking the learners to draft paraphrases based on short texts, incorporate source information into a text through paraphrasing, and to identify and analyse correct and incorrect examples of documenting an original source text. The participants' own drafts produced during Time 1 assessments were also referenced to examine if and how the use of source texts could be improved. Again, because the enrichment program was individualized according to each learner's ZPD, some examples, materials, and exercises were only reviewed briefly or were skipped altogether with certain learners but served as the basis for extensive instruction with others.

The non-enrichment group met with the mediator individually for an equal amount of time as the enrichment group, but they received standardized writing instruction that was not informed by the initial DA diagnoses. However, the instructional materials and activities provided in the non-enrichment program all focused on academic reading and writing and were designed by the researchers in consultation with the English academic writing instructors at the university where the study took place. An example of the activities completed by this group is an adapted task from Ferris and Hedgcock (2014). In the activity, each participant in the group first read a passage on a debatable topic – whether lying is always wrong. Next, the student read two essays composed by another two L2 English learners that responded to the following writing prompt after they had read the source reading: 'Is lying always wrong? Why or why not? Be sure to consider both sides of the issues as you explain your opinion. References to the article – facts, quotations, summary, etc. – are required.' After reading both the source article and the students' essays, the participants were instructed to provide commentary on each essay in terms of its content, organization, language, and the use of the source material. They then compared the two essays to analyse their strengths and weaknesses and offered suggestions on how the essays could be improved.

5 Data analysis

To examine the effectiveness of the enrichment program relative to non-enrichment, or standard instruction, in developing English L2 learners' argumentative writing abilities, the essays produced independently by learners at Time 1, Time 2, and the Transfer task were analysed. The analysis focused on the argumentative practices of stance taking and stance support. Stance taking was examined in terms of both establishing a position and engagement with counterarguments. Engagement with counterarguments was identified as a common challenge experienced by the learners in their initial writing. It was found that the difficulty sometimes was related to their struggle with establishing their own position on the topic and maintaining that position throughout the essay. Therefore, essays were analysed to determine whether the writer established a

Table 4. Categories of coding schemes.

Stance taking	Establishing a position	Taking a position Consistency in the position
	Engagement with counterarguments	Including counterarguments Response to counterarguments
Stance support	Types of evidence Number of source ideas Non-transgressive/transgressive intertextuality Content accuracy	

clearly identified position on the controversial topic and maintained it throughout the text. Regarding engagement with counterarguments, both the inclusion of opposing arguments and rebuttals were considered. In terms of stance support, four sub-features were examined:

1. Whether an essay fulfilled the task expectation by drawing upon both the source readings and the writer's own reasons or experiences as supporting evidence;
2. Number of source ideas used;
3. Whether or not an instance of source use conformed to academic conventions, which, following Borg (2009) and Neumann et al. (2019), was termed non-transgressive intertextuality or transgressive intertextuality; and
4. Whether source information or idea was presented accurately (Neumann et al., 2019; Uludag et al., 2019).

Table 4 presents the categories of the coding schemes.

In analysing the number of source ideas used, the coding scheme created by Uludag et al. (2019) was adopted, where any example, fact, or idea cited from the sources was regarded as a source idea. The coding for non-transgressive/transgressive intertextuality followed Neumann et al., 2019, and we added a sub-category that emerged in our data: indirect source use (very slight modifications, with or without citation), to refer to the situation where a student attempted to paraphrase or summarize, but the language was very close to the original language in the source. Both Neumann et al. (2019) and Uludag et al. (2019) coded accuracy of source use to analyze whether a piece of source information or idea was presented accurately. We adopted the two-point scale reported in Neumann et al. (2019) to assign each source use a score of 2 (accurate presentation of source information), 1 (partial accurate source information), or 0 (misrepresents source reading). A description of the coding scheme for the sub-aspects of the use of supporting evidence is summarized in Table 5. To assess the reliability of coding, a second coder with experience in teaching English academic writing was trained and independently coded 50% of the essays. Intercoder reliability (Miles et al., 2019) reached 92.1% for stance taking (establishing a position and engagement with counterarguments), and the intercoder reliability for the use of supporting evidence was 82.9%.

Table 5. Coding on use of supporting evidence.

Code	Description
Types of evidence	Whether the evidence is from source texts or from the writer's own reasons or experiences
Number of source ideas	Number of examples, facts, or ideas taken from source texts
Non-transgressive intertextuality	Direct quotation: direct quotation with quotation marks with citation
	Indirect source use: paraphrased or summarized with explicit citation
	Indirect source use (implicit): paraphrased or summarized with implicit citation
	Indirect source use (no citation necessary): paraphrased or summarized, no citation necessary
Transgressive intertextuality	Direct quotation (no citation): direct quotation with quotation marks without citation
	Verbatim: verbatim copying of three strings or more without quotation marks, with citation of the source information
	Verbatim (no citation): verbatim copying of three strings or more without quotation marks without citation of the source information
	Indirect (no citation): indirect source use (paraphrased or summarized) without citation
	Indirect source use (very slight modifications, with or without citation)
Content accuracy	2: Accurate reflection of source text
	1: Somewhat accurate reflection of source text
	0: Misrepresents source text

Source. Adapted from Neumann et al., 2019; Uludag et al., 2019.

IV Findings

1 Comparisons of enrichment group and non-enrichment group independent writing at Time 1 and Time 2

The reader will recall that a three-step DA procedure was implemented with all the participants prior to (Time 1) and following (Time 2) the instructional programs. The analyses we present here pertain specifically to learner independent writing at both Times 1 and 2, that is, to the drafts they produced in the first of the three-step DA procedures, before their joint review with the mediator and any efforts by the mediator to support learner identification and correction of problems. In this way, the comparisons that follow capture learner's current level of writing development prior to and following their participation in either the enrichment or non-enrichment instructional programs. In each table presented in this section, features of writing regarding research question 1, stance taking, are presented first, followed by those concerning research question 2, use of supporting evidence.

Table 6 presents the evaluation of integrated argumentative writing produced by the two groups at the start of the study and prior to their instructional programs (Time 1). It can be seen from the table that the two groups were comparable with regard to taking a position and maintaining consistency in the position. One or two students in each group

Table 6. Time 1 pre-mediation writing.

		Enrichment group ($n = 6$)	Non-enrichment group ($n = 7$)
Establishing a position	Taking a position	Hassan did not take a position	Basim did not take a position; Amelia wrote a fence-straddling essay, did not take a clear position
	Consistency in the position	Xuan and Ying took a position but were not consistent	Larry and Zamir took a position but were not consistent
Engagement with counterarguments	Including counterarguments	Abdur and Hassan did not include counterarguments	Basim, Amelia and Ahmad did not include counterarguments
	Response to counterarguments	The other four learners included counterarguments but did not respond to them	Larry, Kumi and Zamir included counterarguments but did not respond to them; Only Jiehua included counterarguments and also responded to them
Use of supporting evidence	Types of evidence	All supported thesis with both information from source texts and their own reasons or experiences.	Ahmad did not use information from source texts.
	Number of source ideas	Sum: 24	Sum: 23
		Mean: 24/6 = 4.0	Mean: 23/7 = 3.3
	Non-transgressive intertextuality	Number: 13	Number: 10
		Percentage: 13/24 = 54.17%	Percentage: 10/23 = 43.48%
	Transgressive intertextuality	Number: 11	Number: 13
		Percentage: 11/24 = 45.83%	Percentage: 13/23 = 56.52%
	Content accuracy	Actual accuracy score: 36	Actual accuracy score: 35
		Mean accuracy score: 36*2/(24*2) = 1.50/2	Mean accuracy score: 35*2/(23*2) = 1.52/2

either did not take a stance as required by the writing task, or they shifted their position in either the body or the conclusion of the text. The two groups were also similar regarding engagement with counterarguments. Two learners in the enrichment group and three in the non-enrichment group did not include any counterarguments in their texts. Only one learner in the non-enrichment group, Jiehua, both included counterarguments and responded to them, whereas none of the learners in the enrichment group did so. In the category of use of supporting evidence, again, the two groups were similar except that the enrichment group gained a higher percentage of non-transgressive intertextuality (and thus a lower percentage of transgressive intertextuality) than the non-enrichment group. This is due in large part to the performance of one enrichment group learner, Hassan, who used four source ideas and paraphrased them all, so the instances of the source use were all coded as non-transgressive. However, the source ideas were not presented accurately in the essay and therefore Hassan gained a low accuracy score.

Following their 5-week writing instruction experience, the participants composed another integrated argumentative essay independently prior to DA interaction at Time 2. Evaluation of the essays produced by the two groups is presented in Table 7. The table shows that the two groups differed in all three categories. First, all the enrichment learners took a position and were consistent in their stance throughout the texts. In the non-enrichment group, three learners did not take a stance, and two learners did not maintain their position in their essays. Second, in the enrichment group, two learners did not include counterarguments, but all the other learners both included counterarguments and responded to them. In the non-enrichment group, on the other hand, four learners did not include counterarguments, and only one learner both included opposing views and responded to them. Lastly, regarding the use of supporting evidence, all the enrichment group learners supported their stance with both information from the sources and their own reasons or experiences, as expected by the writing prompt. In contrast, two learners in the non-enrichment relied solely on source texts and thus wrote a summary of the points selected from the readings, while another two learners from the same group did not use any source information at all. Although the non-enrichment group on average used more source ideas than the enrichment group (4.1 versus 3.2), a noticeably higher percentage of non-transgressive intertextuality was obtained by the enrichment group (68.42% versus 51.72%). The mean accuracy score achieved by the enrichment group (1.78/2) was also notably higher than that gained by the non-enrichment group (1.42/2).

2 Comparisons of enrichment group and non-enrichment group independent writing at transfer assessment

Turning now to the transfer assessment, the reader will recall that the purpose of such a procedure in DA is to ascertain how learners respond when new and more complex tasks are presented and specifically whether learners are able to maintain any developmental gains reached through instruction. In the present study, as explained, the transfer assessment presented learners with three rather than two sources and the readings were at a more difficult level as indicated by their Flesch readability scores.

Table 8 indicates that the two groups performed equally well in establishing a position. Only one learner in each group did not maintain consistency in their stance in text

Table 7. Time 2 pre-mediation writing.

		Enrichment group ($n = 6$)	Non-enrichment group ($n = 7$)
Establishing a position	Taking a position	All took a position	Zamir delayed thesis but did not explain the position clearly; Jiehua did not take a position on the topic; Amelia was ambivalent in her position
	Consistency in the position	All maintained consistent position	Basim and Amelia were inconsistent in arguments
Engagement with counterarguments	Including counterarguments	Xuan and Hina did not include counterarguments	Jiehua, Ahmad, Amelia, and Larry did not include counterarguments
	Response to counterarguments	The other learners all included counterarguments and also responded to them	Zamir and Basim included counterarguments but did not respond to them; Only Kumi included counterarguments and also responded to them
Use of supporting evidence	Types of evidence	All supported thesis with both information from source texts and their own reasons or experiences.	Basim and Jiehua relied solely on source texts whereas Ahmad and Larry did not use information from source texts
	Number of source ideas	Sum: 19 Mean: 19/6 = 3.2	Sum: 29 Mean: 29/7 = 4.1
	Non-transgressive intertextuality	Number: 13 Percentage: 13/19 = 68.42%	Number: 15 Percentage: 15/29 = 51.72%
	Transgressive intertextuality	Number: 6 Percentage: 6/19 = 31.58%	Number: 14 Percentage: 14/29 = 48.28%
	Content accuracy	Actual accuracy score: 34 Mean accuracy score: 34*2/(19*2) = 1.79/2	Actual accuracy score: 41 Mean accuracy score: 41*2/(29*2) = 1.41/2

Table 8. Transfer assessment.

		Enrichment group ($n = 6$)	Non-enrichment group ($n = 7$)
Establishing a position	Taking a position	All took a position	All took a position
	Consistency in the position	Abdur was not consistent	Amelia was not consistent
Engagement with counterarguments	Including counterarguments	Only Xuan did not include counterarguments	Basim, Zamir and Larry did not include counterarguments
	Response to counterarguments	Abdur included counterargument but did not respond to it; the others all included counterarguments and also responded to them	The others included counterarguments and also responded to them
Use of supporting evidence	Types of evidence	All supported thesis with both information from source texts and their own reasons or experiences	All supported thesis with both information from source texts and their own reasons or experiences.
	Number of source ideas	Sum: 19	Sum: 22
		Mean: 19/6 = 3.2	Mean: 22/7 = 3.1
	Non-transgressive intertextuality	Number: 13	Number: 11
		Percentage: 13/19 = 68.42%	Percentage: 11/22 = 50.00%
	Transgressive intertextuality	Number: 6	Number: 11
		Percentage: 6/19 = 31.58%	Percentage: 11/22 = 50.00%
	Content accuracy	Actual accuracy score: 35	Actual accuracy score: 33
		Mean accuracy score: 35*2/(19*2) = 1.84/2	Mean accuracy score: 33*2/(22*2) = 1.50/2

development. Regarding engagement with counterarguments, both groups made improvements compared with their performance at Time 1 and Time 2 assessments. However, the enrichment group outperformed the non-enrichment group in that, with the exception of Xuan, all the other enrichment learners included counterarguments whereas three out of the seven learners in the non-enrichment group did not do so. With regard to the use of supporting evidence, both groups fulfilled the task expectation and used both source readings and their own reasons or experiences as supporting evidence. Meanwhile, the percentage of non-transgressive intertextuality achieved by the enrichment group (68.4%) is again markedly higher than that gained by the non-enrichment group (50%). In the last sub-category, the enrichment group earned notably higher mean accuracy score (1.84/2) than the non-enrichment group did (1.50/2) as well.

V Discussion

Among the 108 articles on L2 argumentative writing reviewed by Farsani et al. (2025), very few focused on the actual quality of argumentation and quite often, argumentative essay was used in a study merely as the chosen genre while the research examined other writing-related topics. In addition, the studies adopting Vygotskian theory tend to focus on collaborative writing in English as a second language (ESL) or English as a foreign language (EFL) contexts, rather than learners' development in L2 argumentative writing abilities over time as impacted by instructional mediation. In the present study, DA served to provide a diagnosis of learner argumentative writing development that included, in SCT terms, abilities that had fully formed and could be apprehended by observing learner independent writing as well as abilities that were still emerging and that came to light through their responsiveness to mediation. Following Vygotsky's (1998) argument that instruction can most effectively promote development when it targets those abilities (i.e., the ZPD), DA diagnoses were used to inform individualized enrichment. Following five weeks of instruction, a parallel version of the initial DA procedure was repeated in order to trace learner development over time. While the use of control groups has not been a common feature in L2 DA research to date, our design included a group of learners who engaged in the initial and subsequent DA sessions but whose instruction was not focused on their ZPD and instead followed a general or standard approach to teaching argumentative writing. The inclusion of control groups was necessary to examine the effectiveness of ZPD-sensitive enrichment relative to other instruction. Finally, a transfer assessment involving more complex tasks than those used during DA was included for the purpose of studying how well participants maintained their argumentative writing abilities as new challenges were introduced.

As the reader will recall, the two research questions motivating this study were to understand the effectiveness of the enrichment program in promoting L2 writers' development of stance taking (i.e. establishing an own position and engagement with opposing positions) as well as their integration of information from source materials and their own reasoning in supporting their stance. Overall, Tables 6 and 7 revealed that whereas the two groups were similar at the start of the study in stance taking and stance support, enrichment learners outperformed the non-enrichment group following the 5-week instructional programs. In the essay writing framework proposed by

Wingate (2012), establishing one's own position is identified as 'the core of argumentation' (p. 152). In the independent writing task completed at Time 2, enrichment participants were more explicit and consistent in establishing their position than the non-enrichment group. Relatedly, in developing their own arguments, the enrichment group was more successful in addressing alternative points of view. In the L2 writing literature, acknowledging alternative points of view and responding to potential objections are seen as writers' ability to engage with readers and to strengthen their relationship with the audience in texts, which is considered central to successful academic persuasion (e.g. Hyland, 2005; Jiang & Hyland, 2025; Jiang & Ma, 2018).

Regarding stance support, all the enrichment learners fulfilled the task expectation by drawing upon both source materials and their own reasoning as supporting evidence. In contrast, half of the learners in the non-enrichment group either merely summarized the source texts or did not integrate any information from the sources at all. This phenomenon was similar to what was reported in Zhang (2013) concerning the pre-test essays produced by the ESL students learning synthesis writing, where they either heavily relied on the sources or just neglected them altogether. Groom (2000) regarded constructing a balance between self and sources, or between the writer's own voice and the voices of others in a field of study, as a defining feature of successful argumentative writing. In this sense, the essays composed by the enrichment learners in Time 2 independent writing in general achieved such a balance, whereas the non-enrichment group continued to have issues to various extents.

With regard to the quality of source use, the enrichment group at Time 2 achieved markedly higher percentage of non-transgressive intertextuality and higher accuracy score than the non-enrichment group did. On the other hand, the non-enrichment group used source ideas much more frequently than the enrichment participants did at Time 2 independent writing. Both Chuang and Yan (2023) and Uludag et al. (2019) note that overreliance on sources may connect with weak argumentation and lack of original content, as 'integrated argumentative writing is not about incorporating as much information from the source texts as possible; it is about creating well-developed arguments with the aid of the provided source materials' (Chuang & Yan, 2023, p. 7).

Regarding the two groups' performance during the transfer assessment, it can be seen from Table 8 that both groups were generally successful in taking a position and being consistent in the position. Both groups also did well in drawing upon source readings as well as their own reasons or experiences in supporting the arguments, as required by the writing prompt. This is a notable achievement considering that the transfer task was more challenging than the previous two writing tasks. For the enrichment group, the learners maintained the progress observed in their performance at Time 2. Improvements observed among the non-enrichment participants between their Time 2 and transfer assessments may be due to the mediation provided to them during the second DA session at Time 2. If so, this could indicate a potential microgenesis, that is, a developmental gain over a very short period of time, which has been discussed in previous DA research (Poehner, 2008). On the whole, however, the enrichment group at the transfer assessment were markedly more successful than the non-enrichment group in the other areas of argumentative writing. Compared with the non-enrichment learners, the enrichment group engaged with the counterarguments more actively, used the source materials more

appropriately, and were also more accurate in presenting source ideas. It is likely that these aspects are relatively challenging for English L2 writers and thus would take more extended mediation, such as the instruction provided in the enrichment program, for them to demonstrate more noticeable improvement.

VI Conclusions

Outside the L2 field, Haywood and Lidz (2007) speculated that one reason DA has not become more mainstream is that its results are often not tightly connected to subsequent instruction, meaning that the use of insights into learner abilities to guide teaching has not been well studied. The research reported here offers an initial step toward exploring this issue in the context of L2 academic writing development. Gains made by learners in the enrichment group relative to their peers in the non-enrichment group provide encouraging results concerning the effectiveness of tailoring instruction to a learner's ZPD. To be sure, this is only a single study involving a small number of participants and with a focus on a particular set of L2 abilities. In that regard, much more research is needed to understand how DA procedures can yield information about learner abilities, how these can be used by teachers and learners, and any differences such efforts might make in promoting learner development.

One important area of continued study will be to expand the focus of assessment and instruction to include examination of learner reading and comprehension of academic texts, which of course connects to the source-based writing focus of the present study (e.g. Plakans, 2009). Moreover, given that academic contexts typically require students to synthesize information presented to them not only through written texts but also aurally, it would be worthwhile for future research to examine implementing DA and DA-informed instruction in enhancing students' abilities in performing integrated reading-listening-writing tasks or video-mediated integrated writing (Machili et al., 2020; Yang & Plakans, 2012).

In addition, future research could consider organizing instructional enrichment according to principles of Vygotskian concept-based language instruction (C-BLI) (Gal'perin, 1992). C-BLI focuses on facilitating learners' internalization of theoretical concepts through systematic instruction mediated by materialized forms such as diagrams, visual modals and flow charts. While C-BLI has emerged in recent years as a major strand of SCT-L2 research (Lantolf et al., 2021), its application to the instruction of L2 writing remains under-researched (although for an exception, see Fu & Liu, 2022). Pairing C-BLI with DA has the potential to provide a powerful way of organizing L2 writing education to promote learner development.

One limitation of the current study concerns the small number of learners recruited to take part in the research, as the study adopted the interactionist approach to DA. Compared with the interventionist approach where an inventory of predetermined mediational moves was delivered to learners during the DA procedures, interactionist DA is more time consuming and labor-intensive, whereas it is also more sensitive to a learner's ZPD and can flexibly adjust the mediation provided depending on learner responsiveness (Lantolf & Poehner, 2004). Existing research has shown that in assessing a genre as complex as argumentation, interactionist DA is more effective than the interventionist

approach in arriving at a more nuanced understanding of learner abilities and improving writing quality (Kushki, Nassaji, & Rahimi, 2022; Nassaji et al., 2020). Future research can explore whether interventionist DA or computerized DA (e.g. Randall & Urbanski, 2023) can be employed when a different genre of writing is examined, or when linguistic aspects of the texts become the focus of research. When larger samples of participants or different focus of inquiries are involved, research methodology can also be expanded to perhaps include principled mixed-methods designs to examine learner writing abilities and textual qualities (Riazi & Farsani, 2024).

Relatedly, the instruction provided to both enrichment and non-enrichment learners occurred on an individualized basis in our study. While this approach allowed for careful tracking of the experiences of individuals and use of diagnostic information to interpret developmental gains, one-to-one teaching is generally not an affordance in most academic writing programs. Future research will wish to examine how ZPD diagnoses might be used to place students in 'dynamic' groups according to similar struggles and similar levels of responsiveness to mediation (Poehner et al., 2018). It is likely that these groups will shift depending upon the focus of instruction, with some learners being grouped together while working on particular topics and then being organized into different groups for other topics. Such an approach has the potential to optimize classroom instruction by engaging students in group activities tailored to their needs and abilities. Finally, rapidly developing AI-powered tools may hold much potential to provide personalized responses to students' writing, functioning perhaps as mediational tools that may transform the process of learning to write (e.g. Barrot, 2023; Warschauer et al., 2023). Future research may wish to explore how AI-enhanced DA could allow for assessment procedures that can be administered to larger groups of learners to diagnose and promote their writing development. This will enable AI technologies to be integrated more fully into the instruction and assessment of academic argumentative writing.

Author contributions

Equal contributions by authors. Author 1 was primarily responsible for data collection and data analysis; author 2 aided in data analysis and study design, and both authors contributed equally to preparation of this manuscript.

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This study was approved by the Penn State Institutional Review Board, and was determined that the study did not require formal Institutional Review Board review because the research met the criteria for exempt research according to the policies of Penn State and the provisions of applicable federal regulations.

Consent to participate

Written informed consent to participate in the study was obtained from the participants.

Consent for publication

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